

NYC 311 Resolution Time Analysis

First Three Months of COVID-19 in New York City

Executive Summary • Rogue Lafayette

The Question

How did the early COVID-19 pandemic affect the resolution times of high-volume 311 service complaints across New York City, and what does that reveal about how city operations held up under crisis?

Approach

Extracted ~490,000 NYC 311 service requests from Google BigQuery covering March 1 to June 1, 2020 — the first three months of confirmed COVID-19 in NYC. After removing 6,535 records with invalid resolution times, the analysis filtered to complaint types with 10,000+ requests and examined resolution times by complaint type and sub-category (descriptor).

Key Finding

Two complaint types appeared slow on average — but for completely different reasons. Distinguishing genuine delay from outlier distortion changed the entire conclusion.

Complaint Type	Mean	Median	Verdict
Unsanitary Condition	35.4 days	20.2 days	Genuinely slow
Consumer Complaint	15.6 days	1.7 days	Outlier-skewed

- Unsanitary Conditions were systemically slow — half of all cases took 20+ days, and nearly 40% exceeded 30 days. Sewage, pests, and mold drove the backlog, all requiring in-person inspection disrupted by lockdown.
- Consumer Complaints only appeared slow — the 15.6-day average was inflated by extreme outliers (one rental hall case took 210 days). The median of under 2 days reflects typical performance.
- The pattern held across the dataset: complaints requiring physical inspection lagged, while remotely-handled complaints (noise, parking) resolved same-day — consistent with reduced field operations during early COVID.

Why It Matters

Reporting the average alone would have mischaracterized consumer complaint performance as poor when it was fast for most cases. This reinforces a core analytical principle: when outliers are present, compare mean against median before drawing conclusions. The two metrics tell different operational stories that would drive different decisions — one category needs process improvement, the other needs outlier investigation.

Caveats

- Findings are correlational, not causal - a pre-COVID baseline would strengthen the conclusion.
- Analysis covers only the first three months; longer timeframes may reveal recovery patterns.

Tech stack: Python (pandas, matplotlib, seaborn) • Google BigQuery • SQL • Google Colab